

Out-of-Distribution Detection Inverts Under Domain-Adversarial Training

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Abstract

Background. Domain-adversarial invariance training is widely recommended for removing acquisition shortcuts from medical imaging models. Its effect on the out-of-distribution (OOD) monitoring that such models rely on at deployment has not been measured.

Methods. We swept the adversarial weight λ_{adv} across seven values with three to five seeds each (27 runs) on ISIC 2019 (dermoscopy, 25,331 images) and PAD-UFES-20 (clinical photographs, 2,298 images), holding everything else fixed. At every λ_{adv} we measured domain leakage by linear probe, in-distribution balanced accuracy, calibration on in-distribution and out-of-domain data, OOD detection with five scoring rules, and zero-shot cross-domain diagnostic accuracy on a patient-level held-out PAD partition. We repeated the sweep with plain single-encoder adversarial training and with four non-adversarial invariance objectives, and validated on Fitzpatrick17k, a third dataset never seen by the adversary.

Results. At the smallest non-zero weight tested, OOD detection fell from AUROC 0.843 to 0.513; by $\lambda_{\text{adv}} = 2$ it had **inverted below chance** to 0.427 (95% CI [0.414, 0.439]; all five seeds individually below 0.5), ranking images from a clinic the model had never seen as more in-distribution than its own test data, throughout the score distribution rather than only in the tails. All five scoring rules collapsed together, the inversion held on the third dataset (0.391 at $\lambda_{\text{adv}} = 2$ against 0.764 at $\lambda_{\text{adv}} = 0$, the same model and data) and survived shrinkage estimation. The collapse was a cliff rather than a trade-off: no operating point combined invariance with detection. It was invisible to routine monitoring — domain leakage fell to its chance floor ($0.915 \rightarrow 0.553$, floor 0.5) and in-distribution balanced accuracy did not degrade ($0.692 \rightarrow 0.707$). Confidence on never-seen-domain images rose from 0.489 to 0.930, overtaking in-distribution confidence, while cross-domain accuracy did not improve at any λ_{adv} and predictions collapsed toward the benign majority class (actinic keratosis recall $0.244 \rightarrow 0.016$, $n = 202$). The failure reproduced in plain single-encoder gradient-reversal training but not in moment-matching objectives, which reduced leakage comparably without collapsing any detector. In a same-modality shift between two hospitals the collapse reproduced (k -NN $0.658 \rightarrow 0.486$) but the inversion did not: with matched baseline detectability, Mahalanobis fell only to 0.683. Six candidate explanations were tested and refuted, and a pre-registered synthetic reproduction failed.

Conclusions. Across imaging modality, domain-adversarial training does not merely weaken out-of-distribution monitoring; it reverses it, while every routinely monitored metric improves. Between two hospitals sharing a modality the collapse persists and the reversal does not, so the two should be claimed separately. OOD detection and out-of-domain calibration should be reported as a function of achieved invariance rather than at a single operating point.

Keywords: domain-adversarial training; out-of-distribution detection; calibration; distribution shift; skin lesion classification; model monitoring

1. Introduction

Deep models for skin lesion classification are known to exploit acquisition artefacts rather than pathology: ruler marks, ink annotations, imaging hardware and site identity all correlate with diagnosis in public benchmarks (Winkler et al., 2019; Bissoto et al., 2020; Daneshjou et al., 2022). The standard remedy is to make the representation invariant to the acquisition domain, most often through domain-adversarial training with a gradient reversal layer (Ganin et al., 2016; Bousmalis et al., 2016). The stated purpose of this remedy is safe deployment across sites.

Deployment across sites also relies on a second mechanism: out-of-distribution monitoring. A deployed system is expected to flag inputs unlike its training data so that a human reviews them rather than the model acting alone. This paper asks what invariance training does to that mechanism.

The answer is that it reverses it. Sweeping the adversarial weight in a controlled setting, we find that out-of-distribution detection does not merely weaken as domain information is removed — it passes through chance and inverts. At the weights that achieve invariance, a model asked which inputs are unlike its training data answers with its own test split, and ranks images from a clinic it has never seen as the more routine of the two. AUROC falls from 0.843 to 0.427, with a 95% confidence interval of [0.414, 0.439] that lies entirely below chance, across every scoring rule we tried.

Degradation would be unsurprising. A representation stripped of domain information should be worse at recognising a domain, and chance is what that argument predicts. Inversion is not predicted by it, and after testing and refuting six explanations and one pre-registered synthetic reproduction, we cannot account for it.

What makes the reversal consequential rather than merely odd is that nothing in a standard evaluation reveals it. Over the same range the model becomes more confident on unseen-domain images than on its own held-out test data, and its predictions collapse toward the benign majority class. Meanwhile in-distribution accuracy does not fall, in-distribution calibration does not change, and the domain leakage probe — the very metric the technique is adopted to improve — improves exactly as intended.

The technique is adopted for safe deployment across sites, and it disables the cross-site safety machinery while every monitored number moves in the direction its user is hoping for.

Contributions.

1. A controlled intervention: a seven-point sweep of the adversarial weight, 27 training runs with everything else held fixed, isolating invariance pressure from architectural and data confounds.
2. The inversion: OOD detection passes through chance and reverses, as a cliff rather than a trade-off curve, across five scoring rules, on a third dataset the adversary never saw, and robustly to shrinkage estimation and tail truncation.
3. The invisibility: in-distribution accuracy, in-distribution calibration and the domain leakage probe all fail to reveal it, and the leakage probe does not mediate it, so it cannot serve as a proxy.

4. Scope: the failure reproduces in plain single-encoder adversarial training, so it is not an artefact of the factorized architecture; and it does not appear under moment-matching objectives that reduce leakage along the same axis, so it belongs to adversarial optimisation rather than to invariance.
5. Six candidate explanations tested and refuted, plus a pre-registered synthetic reproduction that failed — delimiting the phenomenon for whoever explains it.
6. A reporting protocol that makes the failure visible at near-zero cost.

2. Related work

Shortcut learning in dermatology. Winkler et al. (Winkler et al., 2019) showed that ink markings shift melanoma predictions; Bissoto et al. (Bissoto et al., 2020) showed that classifiers retain competitive accuracy on ISIC data when the lesion itself is masked; Daneshjou et al. (Daneshjou et al., 2022) documented degraded performance on darker skin tones. These motivate invariance-based remedies.

Invariant representation learning. Domain-adversarial training (Ganin et al., 2016) and domain separation networks (Bousmalis et al., 2016) remove domain information by adversarial or orthogonality constraints. **The architecture used here is equivalent to a domain separation network and is not a contribution of this paper.** It is a vehicle for the intervention.

Covariate shift versus semantic novelty. Full-spectrum OOD detection (Yang et al., 2023) argues that the OOD literature conflates two distinct shifts, and proposes a four-way taxonomy: training ID, *covariate-shifted ID*, near-OOD and far-OOD. Crucially, it prescribes that covariate-shifted in-distribution samples should be **tolerated as in-distribution rather than flagged**, on the grounds that rejecting them contradicts the goal of generalization. OpenOOD v1.5 (Zhang et al., 2023) adopts the same full-spectrum setting at ImageNet scale and for foundation models.

Our setting falls squarely in the covariate-shifted-ID cell: PAD-UFES-20 contains the same diagnostic classes as ISIC, photographed differently. A reader of (Yang et al., 2023) might therefore regard a model that accepts these images as in-distribution as behaving *correctly*.

We argue this is exactly the gap. **Tolerance of covariate shift is safe only when accompanied by generalization, and the two are evaluated separately or not at all.** Our models achieve the prescribed tolerance — by $\lambda_{\text{adv}} = 2$ they rank out-of-domain images as more in-distribution than their own test data — while cross-domain balanced accuracy sits at 0.249 against a floor of 0.167, plain accuracy is below the majority-class rate, and confidence on those images exceeds in-distribution confidence. Tolerating inputs one cannot classify is worse than flagging them, and no current protocol separates the two cases. We therefore report acceptance and cross-domain accuracy jointly throughout.

Calibration under adaptation. Ovadia et al. (Ovadia et al., 2019) showed that calibration degrades under shift for ordinary training, and Wang et al. (Wang et al., 2020)

established the sharper result that domain adaptation methods specifically *trade* calibration for accuracy: adapted models are miscalibrated on the target domain, and they propose a post-hoc correction. **The calibration degradation we report in Section 4.4 is therefore not a new phenomenon.** What is new here is its dose-dependence on the adversarial weight, its invisibility to in-distribution calibration, and — separately from calibration altogether — the collapse and inversion of out-of-distribution detection, which a post-hoc confidence correction does not address.

Calibration as a route to invariance — and the reverse. Wald et al. (Wald et al., 2021) established the forward direction: enforcing *multi-domain calibration* yields invariance, with models so calibrated provably free of certain spurious correlations, and they propose a calibration loss (CLOvE) validated on WILDS.

Our results show the reverse implication does not hold. Obtaining invariance adversarially does not yield calibration; it *destroys* out-of-domain calibration in a dose-dependent way while leaving in-distribution calibration untouched. Calibration and adversarial invariance are therefore not interchangeable routes to the same representation, and the direction of enforcement matters.

What is new here. (i) The *shape* of the dependence — a cliff at the smallest effective adversarial weight rather than a graded trade-off; (ii) *sub-chance* inversion, in which out-of-domain data is ranked as more typical than in-distribution data throughout the score distribution; (iii) the demonstration that the failure is invisible to every routinely monitored metric; (iv) the finding that the standard domain leakage probe neither mediates nor predicts it; (v) the consequence for calibration and for diagnostic decisions on out-of-domain data; and (vi) the observation that full-spectrum tolerance is achieved here without the generalization that would make it safe.

3. Materials and methods

3.1. Data

ISIC 2019 (Tschandl et al., 2018): 25,331 dermoscopic images, eight diagnostic classes (MEL, NV, BCC, AK, BKL, DF, VASC, SCC), CC BY-NC 4.0. **PAD-UFES-20** (Pacheco et al., 2020): 2,298 smartphone clinical photographs, six classes mapping into the same label space (DF and VASC absent), CC BY 4.0. **Fitzpatrick17k** (Groh et al., 2021): 3,887 of 16,577 images obtained; annotations CC BY-NC-SA 3.0, images web-scraped under mixed copyright, used for evaluation only and not redistributed.

Images were cropped to a soft lesion-centred bounding box via Otsu thresholding on the deviation from the border median, expanded by a 0.30 margin with a 0.70 minimum crop ratio, and resized to 224×224 .

3.2. Splits and the PAD held-out set

ISIC was split 60/20/20 into train/validation/test, stratified by label (`random_state = 42`).

PAD appears in training and would otherwise also be the OOD set. The classification loss never sees PAD labels (`ignore_index = -1`), but PAD images enter the context and adversarial branches. To remove the resulting circularity we partitioned PAD

at patient level into `pad_adv` (1,582 images, 961 patients), visible to those branches during training, and `pad_heldout` (716 images, 412 patients), never seen in any form. No patient and no lesion appears in both. **All OOD and cross-domain numbers reported in the main text use `pad_heldout`.** An earlier draft of this work reported 0.413 at $\lambda_{\text{adv}} = 2$; that value was scored on `pad_full`, which includes the 1,582 images the adversarial branch saw during training. Fitzpatrick17k provides a second never-adversarial domain.

3.3. Architecture and objective

A dual-encoder model with EfficientNet-B3 backbones produces a lesion latent $z_\ell \in \mathbb{R}^{16}$ (batch-normalised, denoted z_ℓ^{norm}) and a context latent $z_c \in \mathbb{R}^{64}$. Diagnosis uses z_ℓ^{norm} only, through a linear head. The objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{ctx}} \mathcal{L}_{\text{ctx}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} + \lambda_{\text{orth}} \mathcal{L}_{\text{orth}}, \quad (1)$$

with $\mathcal{L}_{\text{cls}}$ on ISIC only, $\mathcal{L}_{\text{ctx}}$ a domain cross-entropy on z_c , $\mathcal{L}_{\text{adv}}$ a domain cross-entropy on z_ℓ^{norm} behind a gradient reversal layer, and $\mathcal{L}_{\text{orth}}$ a squared-cosine penalty between the two latents ($\lambda_{\text{orth}} = 1$ throughout).

AdamW, base learning rate 10^{-4} , adversary learning rate multiplier 30, classifier multiplier 0.2, 40 epochs, batches balanced across domains, checkpoint selected on validation accuracy.

3.4. The intervention

$\lambda_{\text{adv}} \in \{0, 0.25, 0.5, 1, 2, 4, 8\}$; five seeds at $\lambda_{\text{adv}} \in \{0, 2, 8\}$ and three at the interior points, for 27 runs. Everything else is held fixed, so λ_{adv} is the only manipulated variable.

3.5. Measurements

Domain leakage. Logistic probe on frozen features predicting the source dataset, reported as **balanced accuracy, whose chance floor is 0.5 for this two-domain problem**. Earlier work on this dataset pair reports plain accuracy against a majority floor of 0.688; the two scales must not be compared to each other. Both are stated where relevant.

OOD detection. Maximum softmax probability (Hendrycks and Gimpel, 2017), free energy (Liu et al., 2020), maximum cosine to a class centroid, class-conditional Mahalanobis with shared covariance (Lee et al., 2018), and k -NN distance (Sun et al., 2022) ($k \in \{10, 50, 200\}$). All statistics are fit on the ISIC training split alone. **These five rules are not independent evidence:** the classifier is a linear map from z_ℓ^{norm} , so all five are functions of the same 16-dimensional vector. Independent evidence comes from repeating the analysis on the pre-projection backbone representation (Section 4.5).

Calibration. Expected calibration error, 15 bins, computed separately on ISIC test and on `pad_heldout` restricted to the six shared classes. **Note that out-of-domain ECE here is arithmetically close to mean confidence minus accuracy**, so the confidence, accuracy and ECE results in Section 4.4 are three views of one phenomenon, not three independent findings.

Table 1: The intervention. Mean $\pm$ s.d. across seeds. Leakage is balanced accuracy (chance 0.5); cross-domain accuracy is balanced over the six classes shared with PAD-UFES (chance 0.167). Arrows mark the direction a practitioner would call favourable.

λ_{adv}	n	Leakage $\downarrow$	ID bal. acc $\uparrow$	ID ECE	OOD ECE	OOD AUROC	Fitz. AUROC	X-dom bal. acc
0	5	0.915 ± 0.021	0.692	0.107	0.247	0.843 ± 0.023	0.764 ± 0.038	0.291 ± 0.016
0.25	5	0.553 ± 0.008	0.696	0.101	0.684	0.509 ± 0.023	0.439 ± 0.061	0.291 ± 0.020
0.5	3	0.555 ± 0.029	0.701	0.096	0.724	0.465 ± 0.015	0.449 ± 0.025	0.270 ± 0.034
1	3	0.569 ± 0.010	0.701	0.097	0.733	0.455 ± 0.036	0.457 ± 0.088	0.261 ± 0.016
2	5	0.584 ± 0.027	0.707	0.099	0.746	0.427 ± 0.025	0.391 ± 0.068	0.249 ± 0.030
4	3	0.548 ± 0.035	0.686	0.100	0.726	0.455 ± 0.039	0.431 ± 0.036	0.276 ± 0.025
8	5	0.625 ± 0.045	0.679	0.099	0.713	0.494 ± 0.027	0.474 ± 0.060	0.269 ± 0.023
EffNet-B3 control		0.802	0.683	0.096	—	0.726 ± 0.021	0.649	0.298 ± 0.010
ResNet-50 baseline		0.979	0.655	0.101	—	0.868 ± 0.019	0.994	0.391

Uncertainty. Intervals on AUROC come from a stratified bootstrap with 2,000 resamples, percentile method, fixed in a pre-registration before the values were computed. Each draw resamples **patients** from `pad_heldout` and, independently, images from the ISIC test split, because AUROC depends on both score distributions and resampling only one side understates the interval. Within a draw the same resampled sets are used for every seed. We report both the interval on the five-seed mean and the number of seeds whose individual interval clears 0.5. Fitzpatrick17k carries no patient identifiers, so its bootstrap is over images and its interval is likely anti-conservative.

Cross-domain diagnosis. Zero-shot evaluation on `pad_heldout` over the six shared classes. PAD labels were never used for $\mathcal{L}_{\text{cls}}$.

4. Results

4.1. The intervention

Table 1 reports every measurement as a function of λ_{adv} , the only manipulated variable. Figure 1 shows the four quantities that matter most on a shared axis.

Read across a single row and nothing is wrong. Read down the columns and two of them improve while two collapse.

4.2. Invariance is achieved, and in-distribution metrics do not warn

Domain leakage falls from 0.915 to 0.553 between $\lambda_{\text{adv}} = 0$ and $\lambda_{\text{adv}} = 0.25$, against a chance floor of 0.5. By the smallest non-zero weight tested, the lesion representation is very nearly uninformative about the acquisition source: the intended effect of the method, obtained essentially in full.

It is obtained without an in-distribution cost. Balanced accuracy on the ISIC test split moves from 0.692 to 0.707 over the useful range — we claim no improvement, the difference being small relative to seed variance — and in-distribution expected calibration error is flat at ≈ 0.10 at every λ_{adv} . A practitioner tracking accuracy, in-distribution calibration and the leakage probe would conclude that the method had worked exactly as advertised.

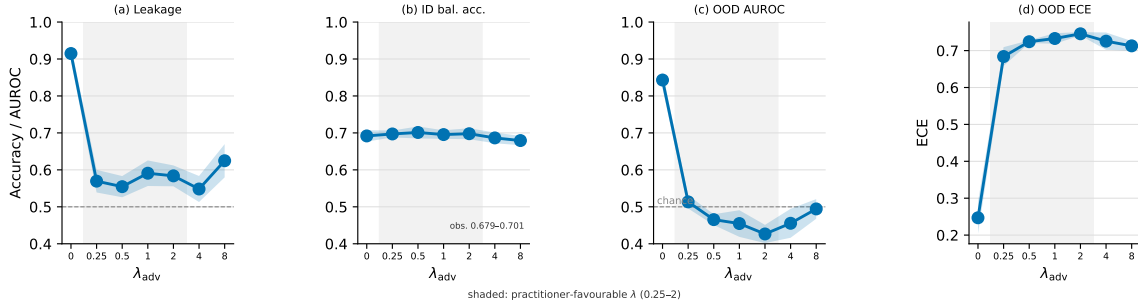


Figure 1: **The intervention, four views, one axis.** Leakage falls to its chance floor and in-distribution balanced accuracy does not degrade — both favourable. Over the same range OOD AUROC collapses through chance and inverts, and out-of-domain calibration error triples. A developer monitoring the first two panels would see nothing wrong.

4.3. Out-of-distribution detection collapses, and inverts

Over the same interval, Mahalanobis AUROC against the held-out PAD partition falls from 0.843 to 0.513, and reaches 0.427 at $\lambda_{adv} = 2$.

Two features of this collapse matter more than its size.

The fall to chance is a cliff; the inversion develops after it. Almost the whole descent happens in one step: 0.843 to 0.509 between $\lambda_{adv} = 0$ and $\lambda_{adv} = 0.25$. The sub-chance region is reached only at $\lambda_{adv} \geq 0.5$, after which the value moves between 0.427 and 0.494 with no ordering that would let a practitioner tune for a safe compromise. We therefore distinguish the two: detection collapses to chance at the smallest non-zero weight tested, and inverts under further pressure. There is no operating point on this curve at which invariance has been obtained and detection survives.

It goes below chance. At $\lambda_{adv} = 2$ the detector ranks images from a clinic the model has never seen as *more* in-distribution than images from its own test split. A patient-clustered bootstrap puts the 95% interval at $[0.414, 0.439]$, and all five seeds have individual intervals entirely below 0.5. The same holds for k -NN (0.440, $[0.429, 0.452]$, 5/5 seeds) and, with one seed short of unanimity, for cosine (0.453, $[0.441, 0.464]$, 4/5). This is not a tail artefact. Removing the pooled 90th, 95th or 99th percentile of scores leaves AUROC at 0.412, 0.416 and 0.424 respectively against 0.427 for the full set, and the medians themselves invert (ISIC 13.8 versus `pad_heldout` 10.0). The shift is distributional throughout.

All five scoring rules move together — MSP $0.970 \rightarrow 0.437$, energy $0.983 \rightarrow 0.441$, cosine $0.901 \rightarrow 0.435$, k -NN $0.941 \rightarrow 0.424$ (stable across $k \in \{10, 50, 200\}$). As noted in the methods these are five views of one 16-dimensional representation and are not independent evidence; the independent replication is at the backbone (Section 4.5) and in the single-encoder model (Section 4.7).

The collapse is not a numerical artefact of the covariance estimate. Ledoit–Wolf shrinkage leaves the $\lambda_{adv} = 2$ value unchanged on the held-out partition (0.427 under both).

Across float32 and float64 covariance estimation, with and without shrinkage, and at one, four and eight BLAS threads, the largest deviation in AUROC is 9.4×10^{-4} . Effective dimensionality barely moves across the sweep (participation ratio $4.879 \rightarrow 4.500$ while leakage falls $0.914 \rightarrow 0.553$).

It is not an artefact of an unusually aggressive adversary. The adversary runs at $30\times$ the base learning rate, and retraining at $10\times$ and $100\times$ leaves the inversion in place: 0.418 ± 0.011 at $10\times$ with all three seeds below 0.5, against 0.432 ± 0.024 at $30\times$ and 0.482 ± 0.020 at $100\times$ (one seed at 0.503). Weakening the adversary tenfold deepens the inversion rather than removing it.

It is not confined to the domain the adversary was trained against. On Fitzpatrick17k, absent from every training branch, AUROC falls $0.764 \rightarrow 0.391$ across the same sweep (95% CI $[0.386, 0.396]$, five of five seeds below 0.5; the bootstrap here is over images, since Fitzpatrick17k carries no patient identifiers, so the interval is likely narrower than the true sampling uncertainty). For context, a ResNet-50 baseline reaches 0.994 on these images and the EfficientNet-B3 single-encoder control 0.649; the controlled comparison is the $\lambda_{\text{adv}} = 0$ row, since it differs from $\lambda_{\text{adv}} = 2$ only in the adversarial weight.

4.4. The model becomes confidently wrong on unseen domains

At $\lambda_{\text{adv}} = 0$ the model is appropriately hesitant on unfamiliar data: mean maximum-softmax confidence is 0.915 on the ISIC test split and 0.489 on `pad_heldout`. By $\lambda_{\text{adv}} = 0.25$ that gap has closed (0.913 versus 0.898), and by $\lambda_{\text{adv}} = 2$ it has inverted — the model is *more* confident on images from a clinic it has never seen (0.930) than on its own held-out test set (0.910).

Confidence rises while accuracy falls. Cross-domain diagnostic accuracy does not improve at any λ_{adv} : balanced accuracy spans 0.249–0.291 against a six-class floor of 0.167, no setting exceeds the 0.291 obtained at $\lambda_{\text{adv}} = 0$, and every setting is below the 0.298 of the non-adversarial EfficientNet-B3 control. Plain accuracy falls $0.245 \rightarrow 0.184$, beneath the 0.385 majority-class rate throughout.

Macro AUC is roughly preserved ($0.605 \rightarrow 0.581$), so discrimination survives while the decision rule does not. Predictions collapse toward the benign majority class: nevus predictions rise from 113 to 374 of 716 images, and **actinic keratosis recall falls from 0.244 to 0.016** ($49/202 \rightarrow 3/202$ images, mean over five seeds). Squamous cell carcinoma moves the same way ($0.088 \rightarrow 0.021$, $6/66 \rightarrow 1/66$) but the count is too small to carry a claim on its own, and melanoma ($n = 9$) supports none.

The resulting out-of-domain calibration error triples, from 0.247 to 0.746, while in-distribution ECE stays flat at ≈ 0.10 . That adapted models are miscalibrated on the target domain is known (Wang et al., 2020); what the sweep adds is that the degradation is dose-dependent on the adversarial weight and leaves in-distribution calibration untouched, so no in-distribution measurement reveals it. We stress that confidence, accuracy and calibration here are three views of a single phenomenon rather than three independent results: in this setting out-of-domain ECE is arithmetically close to mean confidence minus accuracy.

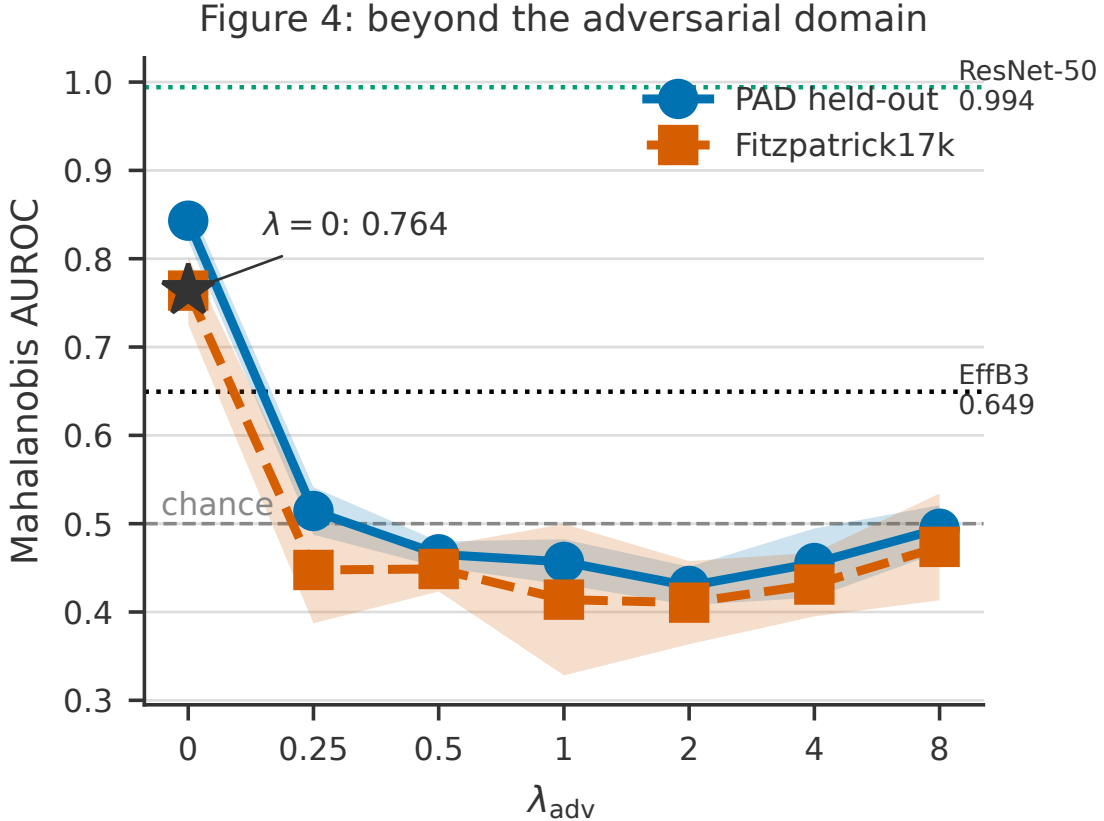


Figure 2: **The collapse is not confined to the adversarial domain.** AUROC against λ_{adv} for the held-out PAD partition and for Fitzpatrick17k, which is absent from every training branch. Horizontal references give the ResNet-50 baseline and the EfficientNet-B3 control on Fitzpatrick; the controlled comparison is the $\lambda_{adv} = 0$ point of the sweep itself.

4.5. Leakage does not mediate the collapse

The natural hypothesis is that removing domain information is what breaks detection. Three results rule it out.

First, leakage **saturates at its chance floor** by $\lambda_{adv} = 0.25$ (0.553 against a floor of 0.5) while AUROC continues to fall from 0.513 to 0.427 as λ_{adv} rises further. The damage continues after invariance is complete.

Second, at the pre-projection backbone, leakage is essentially flat across the whole sweep (0.982 $\rightarrow$ 0.941) while backbone Mahalanobis AUROC falls 0.749 $\rightarrow$ 0.475. Domain information remains linearly decodable there, and detection collapses anyway. Because the backbone is a different representation from the 16-dimensional latent, this is the independent replication that the five-detector agreement does not provide.

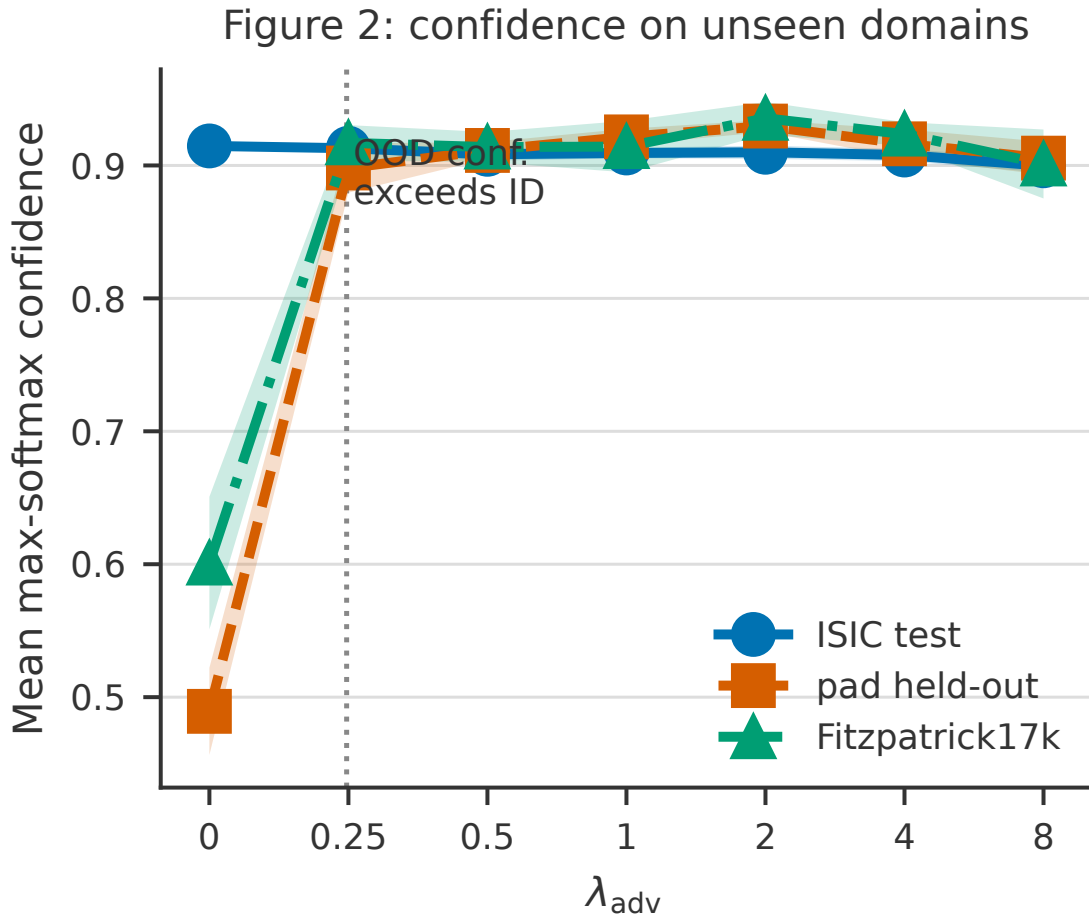


Figure 3: **Confidence on unseen domains overtakes confidence on familiar data.** Mean maximum-softmax probability against λ_{adv} for the ISIC test split, the held-out PAD partition and Fitzpatrick17k. The crossing occurs by $\lambda_{adv} \approx 0.5$. λ spacing is categorical.

Third, and most directly, two training objectives traverse the same leakage range with opposite consequences (Section 4.6). Leakage alone therefore does not determine the outcome.

We accordingly report leakage as a measured quantity and never as the explanatory variable, and plot all results against λ_{adv} rather than against leakage.

4.6. The failure is specific to gradient reversal

Adversarial training is one route to invariance among several. Table 2 sweeps four alternatives on the single-encoder backbone: deep CORAL (Sun and Saenko, 2016), maximum

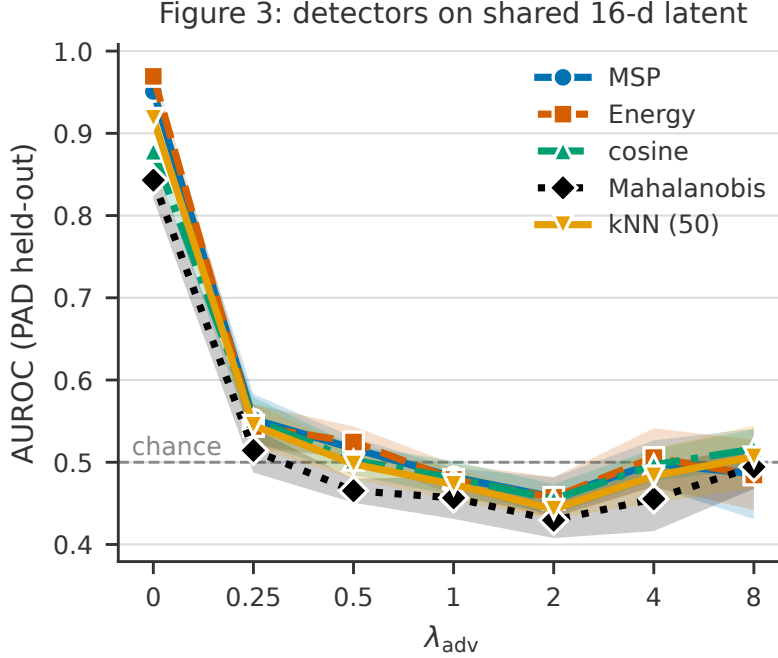


Figure 4: **Every scoring rule moves together.** The five detectors against λ_{adv} on the held-out PAD partition, mean $\pm$ 1 s.d. All five are functions of the same 16-dimensional latent and are therefore not independent evidence; Figure 5 carries the replication on a different representation.

mean discrepancy (Long et al., 2015), invariant risk minimisation (Arjovsky et al., 2019) and group distributionally robust optimisation (Sagawa et al., 2020).

Only MMD moves leakage appreciably, and it does so at a price gradient reversal never pays. Escalating the MMD weight from 1 to 10,000 buys 0.13 of leakage ($0.967 \rightarrow 0.836$, floor 0.5) at a cost of 0.44 of in-distribution balanced accuracy ($0.779 \rightarrow 0.337$), and leaves every detector well above chance (Mahalanobis 0.665, k -NN 0.738, energy 0.694 at the highest weight). IRM reaches the extreme of the same pattern: at weight 100 its balanced accuracy collapses to 0.167 while leakage remains at 0.950. CORAL does not induce invariance at all; at weight 100 it makes the representation *more* domain-separable (leakage 0.998). GroupDRO leaves leakage unmoved at every weight tested.

The contrast is cleanest when task health is held constant rather than leakage, since MMD only reaches comparable leakage by destroying the classifier:

Gradient reversal reduces leakage $0.970 \rightarrow 0.763$ **at no cost in accuracy** ($0.845 \rightarrow 0.845$) and collapses all five detectors to chance. MMD reduces leakage $0.967 \rightarrow 0.900$ at a similarly negligible cost ($0.779 \rightarrow 0.730$) and collapses nothing (Mahalanobis $0.813 \rightarrow 0.716$). To go further, MMD must destroy the task.

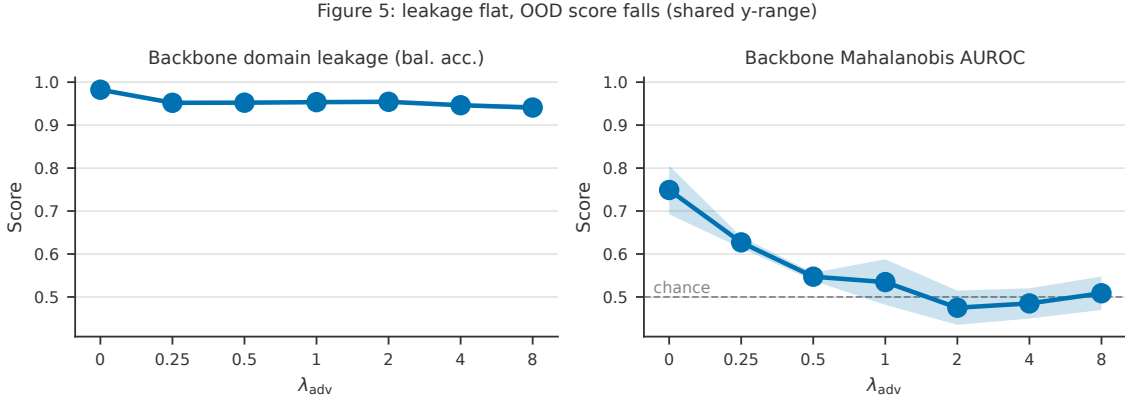


Figure 5: **Leakage does not mediate the collapse.** At the pre-projection backbone the domain probe stays flat across the sweep while Mahalanobis AUROC falls. Domain information remains linearly decodable and detection degrades regardless.

Table 2: Invariance objectives other than gradient reversal, single-encoder backbone, 3 seeds per cell. Leakage is balanced accuracy, floor 0.5. **IRM and GroupDRO use PAD labels** through their environment and group risks; the remaining objectives do not, so their cross-domain numbers are not comparable and the two regimes are kept in separate blocks.

Objective	Weight	Leakage	Maha. AUROC	ID bal. acc	OOD ECE
<i>No access to PAD labels</i>					
ERM [§]	—	0.973	0.838	0.761	—
CORAL	100	0.998	0.999	0.777	—
MMD	1	0.967	0.813	0.779	—
MMD	100	0.900	0.716	0.730	—
MMD	3000	0.893	0.748	0.382	0.142
MMD	10000	0.836	0.665	0.337	0.267
GRL	$\lambda_{adv} = 0.25$	0.763	0.525	0.845[†]	—
<i>Uses PAD labels — not comparable on cross-domain accuracy</i>					
IRM	100	0.950	0.729	0.167	—
GroupDRO	10	0.976	0.868	0.783	—

[†] plain accuracy for the GRL row; see Table 3.

[§] scored on `pad_full` with a uniform optimiser and an ISIC-only loss; the $\lambda_{adv}=0$ row of Table 3 uses `pad_heldout`, grouped learning rates and `ignore_index`. They are not the same condition. [PENDING: harmonise or relabel.]

The calibration signatures differ in the same direction. As MMD degrades, its out-of-domain ECE *falls* (0.344 $\rightarrow$ 0.142): the model becomes less confident as it becomes less accurate, which is the appropriate response. Under gradient reversal out-of-domain ECE

Table 3: Single-encoder DANN on the same data and splits. ID accuracy is plain accuracy; leakage is balanced accuracy, floor 0.5. All five detectors collapse at the smallest non-zero weight, and four of five fall below chance by $\lambda_{\text{adv}} = 4$.

λ_{adv}	n	Leakage	Maha.	k -NN	MSP	cosine	Energy	ID acc	X-dom
0	5	0.970	0.700	0.944	0.868	0.884	0.911	0.845	0.390
0.25	3	0.763	0.525	0.528	0.515	0.509	0.507	0.845	0.333
0.5	3	0.795	0.510	0.517	0.501	0.493	0.507	0.844	0.328
1	3	0.795	0.548	0.521	0.462	0.482	0.448	0.840	0.334
2	5	0.807	0.509	0.516	0.479	0.468	0.470	0.839	0.325
4	3	0.798	0.502	0.486	0.471	0.454	0.449	0.835	0.320
8	3	0.787	0.520	0.532	0.519	0.490	0.500	0.832	0.321

triples while accuracy falls. The two objectives move leakage the same way and fail in opposite manners — one degrades honestly, the other is confidently wrong.

We do not claim MMD would be safe at the invariance level gradient reversal reaches; it never got there. What the data establish is that at the leakage levels MMD does reach, it shows none of the failure, while gradient reversal at the same levels has already collapsed.

4.7. It is not specific to the factorized architecture

All results above use a dual-encoder model. Table 3 repeats the sweep with plain single-encoder domain-adversarial training: one backbone, one linear classifier, a gradient reversal layer and a domain head, with no context branch and no orthogonality term.

The cliff reproduces. Every detector collapses at $\lambda_{\text{adv}} = 0.25$ — k -NN $0.944 \rightarrow 0.528$, energy $0.911 \rightarrow 0.507$, cosine $0.884 \rightarrow 0.509$, MSP $0.868 \rightarrow 0.515$ — and by $\lambda_{\text{adv}} = 4$ four of the five sit below chance, cosine most clearly at 0.454 ± 0.014 . In-distribution accuracy is flat throughout ($0.845 \rightarrow 0.832$) and cross-domain accuracy falls from 0.390 to a plateau near 0.32. The invisibility, the collapse and the inversion are all properties of domain-adversarial training rather than of the factorized architecture.

A second backbone leaves the question open. Repeating the single-encoder sweep with a ResNet-50 backbone did not settle whether the inversion is specific to EfficientNet-B3, because comparable invariance was never reached. At $\lambda_{\text{adv}} = 2$ the adversary never left chance — best-epoch cross-entropy 0.686 to 0.688 against a threshold of 0.673 — so those three runs are null experiments and we draw no conclusion from them. At $\lambda_{\text{adv}} = 0.5$, where the adversary did learn (best-epoch cross-entropy 0.41 to 0.55), leakage fell only from 0.965 to 0.939 against the 0.742 that EfficientNet-B3 reaches on the same partition, and Mahalanobis declined to 0.814 without approaching chance. The backbone question is therefore untested rather than answered in the negative.

That run is nonetheless the sharpest evidence for Section 4.5. With the gate valid and leakage moving by 0.026, k -NN fell from 0.962 to 0.563 and cosine from 0.922 to 0.543. The outcome moves by 0.40 while the supposed mediator moves by 0.026.

One difference is informative. The single-encoder model plateaus at leakage 0.76–0.81 and never reaches the 0.553 the dual-encoder attains; more adversarial pressure buys no

further invariance. The dual-encoder has somewhere to put the discarded information — a domain probe on its context branch reads > 0.9999 — while the single-encoder must destroy it. The architecture that achieves deeper invariance also suffers the larger collapse (0.427 versus 0.509 at $\lambda_{\text{adv}} = 2$), which is the dose-response relationship seen from a second direction.

4.8. The collapse generalises to a same-modality site shift; the inversion does not

Every result above contrasts dermoscopy with clinical photography, so the construct we call domain is confounded with imaging modality. ISIC 2019 contains a contrast that is not: its `lesion_id` field records the contributing site, with 12,413 images from Hospital Clínic Barcelona (BCN) and 10,015 from the HAM10000 collection (HAM). Two hospitals, two dermatoscopes, two patient populations — one modality.

The sites are separable, and not for trivial reasons. A logistic probe on frozen ERM features distinguishes them at balanced accuracy 0.962 ± 0.007 against a floor of 0.5, comparable to the 0.970 that ISIC $\leftrightarrow$ PAD reaches at $\lambda_{\text{adv}} = 0$. Masking the dermatoscope vignette reduces this only to 0.943 ± 0.005 , so the border is not the signal. Hand-crafted colour and intensity statistics reach 0.901 at best — close to the 0.876 ceiling the same features reach on ISIC $\leftrightarrow$ PAD, so this pair is no more low-level than the original contrast — while sharpness statistics sit at chance (0.532). A probe given only the ground-truth class label reaches 0.668, reflecting the two sites’ different class distributions; the feature probe is far above that reference.

The collapse reproduces. Training single-encoder DANN with BCN as the in-distribution set and a lesion-disjoint 30% HAM partition held out, leakage falls from 0.963 ± 0.006 to 0.821 ± 0.030 at $\lambda_{\text{adv}} = 0.25$, and the detectors that have headroom fall through chance: k -NN $0.658 \pm 0.041 \rightarrow 0.486 \pm 0.009$ and cosine $0.576 \pm 0.040 \rightarrow 0.477 \pm 0.006$ (3 seeds). The leakage reduction matches the single-encoder dermatology sweep almost exactly ($0.970 \rightarrow 0.763$), and unlike the two multi-domain settings in the limitations, the manipulation took.

Mahalanobis is uninformative in this direction: it reads 0.379 at $\lambda_{\text{adv}} = 0$, already below chance before any adversarial training. Reweighting the in-distribution set to the out-of-domain class mix raises it to 0.487, so most of the baseline inversion is class composition — HAM is two-thirds nevus and therefore concentrated near a single well-populated centroid — with a residual we attribute to HAM10000’s more standardised acquisition and correspondingly lower feature variance.

With the roles reversed, the inversion does not appear. Taking HAM as the in-distribution set restores the headroom: Mahalanobis reads 0.846 ± 0.003 at $\lambda_{\text{adv}} = 0$, essentially identical to the 0.843 of the dermatology sweep. Adversarial training then reduces leakage from 0.95 to 0.77 and drives Mahalanobis to 0.719 ± 0.019 at $\lambda_{\text{adv}} = 0.25$ and 0.683 ± 0.015 at $\lambda_{\text{adv}} = 1$. The class-reweighted variant falls from 0.74 to 0.54. Detection degrades substantially — and stops above chance: at $\lambda_{\text{adv}} = 1$ the patient-clustered bootstrap gives a 95% interval of [0.666, 0.699], and all three seeds have individual intervals lying entirely above 0.5.

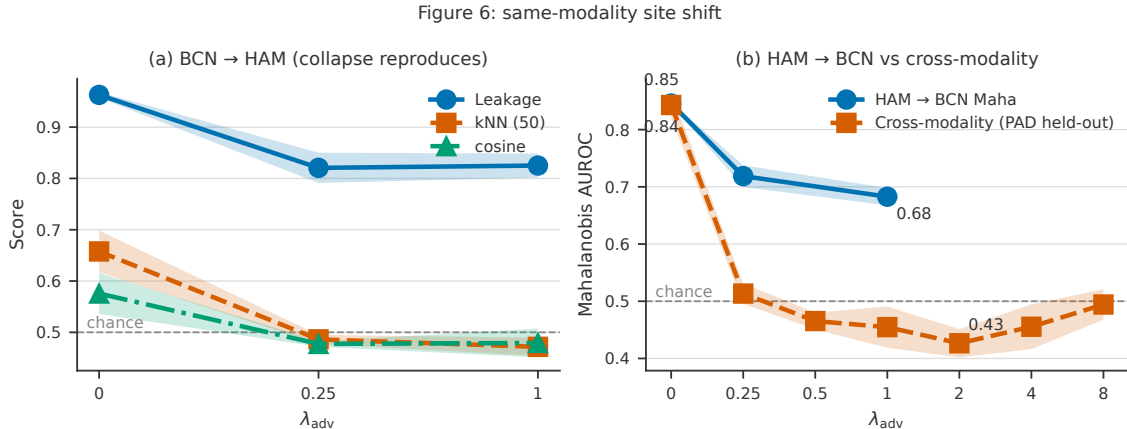


Figure 6: **A same-modality site shift separates the two claims.** (a) Barcelona to HAM10000: leakage falls and the detectors with headroom cross chance. (b) With the roles reversed to restore headroom, Mahalanobis falls from a baseline matching the cross-modality sweep but stops above chance, while the cross-modality sweep continues below it.

Reading. The comparison is tighter than it first appears. Baseline detectability is the same (0.846 versus 0.843) and the achieved invariance is the same as the single-encoder dermatology run (0.77 versus 0.763), yet the cross-modality contrast reaches 0.427 while the same-modality contrast stops at 0.683. Neither a shortage of headroom nor a shortfall in achieved invariance accounts for the difference.

We therefore separate two claims that the earlier sections report together. The *collapse* — detection falling to chance as invariance is obtained, invisibly to in-distribution metrics — reproduces under a same-modality shift between two hospitals. The *inversion* does not: in every setting where we observe it, the two domains differ in imaging modality. Whether that reflects a threshold in the size of the shift, something specific to modality gaps, or a property of these particular datasets, we cannot say.

4.9. What the phenomenon is not

Table 4 lists six candidate accounts, each tested against a prediction fixed in advance, and each refuted, together with a pre-registered synthetic reproduction that failed. We report them prominently because they are the reason the remaining claims are credible, and because delimiting a phenomenon is useful to whoever eventually explains it. We do not have a mechanism.

4.10. A domain monitor is cheap, and semantic novelty is unaffected

A linear domain head on a frozen ImageNet ResNet-50 that has never seen this data reaches AUROC 0.998; on the trained baseline, 0.997; on the trained EfficientNet-B3 control, 0.991. Detecting the acquisition domain is easy from almost any representation, so retaining a

Table 4: Candidate explanations tested and refuted. Each was a prediction specified before its test.

Explanation	Test	Result
Memorisation of the adversarial images	Hold PAD out of the adversary at patient level	<code>pad_heldout</code> 0.427 vs <code>pad_adv</code> 0.407; unchanged
Class composition	Restrict ID to the six shared classes; reweight ID to the OOD class mix	0.432 vs 0.427 on <code>pad_heldout</code> ; reweighting <i>strengthens</i> the inversion to 0.240
Domain specificity	Evaluate on Fitzpatrick17k, never in $\mathcal{L}_{adv}$	0.391 at $\lambda_{adv} = 2$ against 0.764 at $\lambda_{adv} = 0$, same model and data
Dimensional collapse	Participation ratio and eigenspectrum across λ_{adv}	$4.879 \rightarrow 4.500$ while leakage $0.914 \rightarrow 0.553$; decoupled
Latent compression	Latent variance and distance to centroid across λ_{adv}	OOD is already $4.5\times$ tighter at $\lambda_{adv} = 0$; the adversary <i>decompresses</i> it
Tail asymmetry	Recompute AUROC after truncating the pooled upper tail	0.412–0.424 versus 0.427; medians invert
Synthetic reproduction	Pre-registered toy model with heavy-tailed ID structure, grid fixed before fitting	No inversion ($\lambda_{adv}=0$: 0.693; $\lambda_{adv}=2$: 0.786)

monitor is not in itself difficult. What distinguishes the dual-encoder’s context branch is *concentration*: a single principal component carries 84.5% of its variance and already separates the domains (AUROC > 0.9999, two discordant pairs in 11.6M), against 0.55–0.70 at $k = 1$ for every other representation tested.

The damage is specific to covariate shift. With the domain held constant and diagnostic classes withheld from training instead, both branches land between 0.6 and 0.8 and the ordering depends on which class is withheld (DF and VASC: lesion 0.648, context 0.641, baseline 0.687; SCC: lesion 0.806, context 0.607, baseline 0.684). No near/far structure organises the difference (Pearson -0.12 against a centroid-similarity index). We report this as a null result.

5. Discussion

5.1. An inversion, not a degradation

The result we consider central is not that out-of-distribution detection weakens under adversarial invariance training. It is that the ordering reverses. At $\lambda_{adv} = 2$, a model asked which images are unlike its training data answers with its own test split, and ranks images from a clinic it has never seen as the more routine of the two. Every scoring rule we tried agrees, the effect holds on a third dataset the adversary never saw, and it survives shrinkage estimation and truncation of the score distribution’s tails.

Degradation would be unsurprising: a representation stripped of domain information should be less able to recognise a domain. Chance performance is what the obvious argument predicts, and chance is what a single-encoder model reaches at $\lambda_{adv} = 0.25$ (0.525). Going *past* chance is not predicted by that argument, and we cannot currently explain it.

5.2. The deployment pathway

The results compose into a concrete failure sequence. A model is trained with invariance to acquisition domain, as recommended, and deployed at a new site. On images from that site the model increasingly predicts the benign majority class, with actinic keratosis recall falling from 0.244 to 0.016 (49 of 202 images correctly recalled, then 3). It does so with higher confidence than on its own test data. The monitor that should flag those images for human review scores them as more routine than familiar ones — not merely failing to flag them, but actively ranking them as safer. And the developer’s evaluation shows in-distribution accuracy holding, in-distribution calibration unchanged, and the leakage metric improved exactly as intended.

5.3. The failure belongs to gradient reversal, not to invariance

Reducing domain leakage is not sufficient to produce the failure. MMD reduces leakage along the same axis and collapses nothing: at the leakage levels it reaches with the classifier intact, every detector remains well above chance. Gradient reversal at comparable leakage has already inverted.

The two objectives also fail in opposite manners. As MMD is pushed harder it becomes less accurate *and less confident*, and its out-of-domain calibration error falls. Under gradient reversal accuracy falls while confidence rises, and calibration error triples. One degrades honestly; the other is confidently wrong. Whatever drives the inversion is a property of how invariance is obtained, not of invariance itself. We tested one adversarial formulation. Whether other adversarial objectives — CDAN, adversarial discriminative adaptation — behave like gradient reversal or like the moment-matching objectives is not established here, so the claim is stated for gradient reversal rather than for adversarial training in general.

We cannot rule out that MMD would behave the same way at the invariance level gradient reversal reaches, because it never got there without destroying the task. The claim is bounded accordingly.

5.4. The danger zone is where the method succeeds

Risk scales with *achieved* invariance, not with the nominal weight or with the presence of an adversarial term in the loss. In a separate setting we ran the same objective with an adversary that never learned; leakage stayed at 0.94 across four orders of magnitude of λ_{adv} , and the safety failure did not appear either. A weak adversary produces neither debiasing nor harm. Many deployed configurations may be too weak to induce invariance, and are correspondingly harmless — which also means that a configuration’s safety cannot be read off its hyperparameters.

5.5. Tolerance without generalisation

Full-spectrum OOD detection (Yang et al., 2023) prescribes that covariate-shifted in-distribution inputs be accepted rather than flagged. Our models achieve that prescription and are unsafe because of it. By $\lambda_{\text{adv}} = 2$ they accept out-of-domain images more readily than their own test data, while classifying them at 0.249 balanced accuracy against a floor of 0.167 and reporting higher confidence than in distribution.

Acceptance and competence are separate properties, and current protocols evaluate them separately or not at all. We suggest that any claim of tolerance to covariate shift be accompanied by cross-domain accuracy on the same inputs, since tolerating inputs a model cannot classify is a failure mode rather than a success.

5.6. Two routes to invariance are not interchangeable

Wald et al. (Wald et al., 2021) showed that enforcing multi-domain calibration produces invariance. We find the converse fails: obtaining invariance adversarially destroys out-of-domain calibration, driving ECE from 0.247 to 0.746. The property one route uses as its foundation is the property the other route removes.

This suggests a design preference. Where invariance is the goal, calibration-based or moment-matching objectives may be preferable to adversarial ones, because they reach invariance through constraints that leave the out-of-domain uncertainty signal intact. Our MMD results are consistent with this, but we have not tested it directly and offer it as a hypothesis the present results motivate.

5.7. Recommendation

Report OOD detection and out-of-domain calibration **as a function of achieved invariance**, not at a single operating point; report cross-domain accuracy alongside any claim of tolerance to covariate shift; and state, when a separate dataset serves as the OOD set, that this measures covariate shift rather than semantic novelty. Practitioners tuning λ_{adv} already train at multiple values, so this is a change in what is reported rather than additional work. It converts an invisible failure into a visible number.

5.8. An open mechanism

Six accounts were tested and refuted and a pre-registered synthetic reproduction failed. We do not know why the inversion occurs. We know that it is a bulk location shift rather than a tail effect, that it survives shrinkage estimation, that it is not mediated by linear domain-decodability, that it appears on domains the adversary never saw, that it reproduces without the factorized architecture, and that moment-matching objectives do not produce it. We report the eliminations so that the search space is smaller for whoever resolves it.

6. Limitations

Absolute cross-domain performance is poor for every model here. Plain accuracy on `pad_heldout` is 0.245 for the factorized model and 0.391 for the baseline, against a 0.385 majority rate. No model in this study is deployable. Our claim concerns the *effect of* λ_{adv} within a controlled intervention, not the deployability of any particular model, and the deployment pathway above should be read as a mechanism of harm rather than as a description of a fielded system.

The inversion is observed only across imaging modality. ISIC is dermoscopy; PAD-UFES and Fitzpatrick17k are clinical photography, so the main contrast confounds site with modality, and Fitzpatrick17k does not resolve this because it is also clinical photography. The same-modality contrast within ISIC (Section 4.8) separates the two claims:

the collapse reproduces between two hospitals, the inversion does not. Every setting in which we observe the inversion is a cross-modality one, and we cannot say whether that reflects the size of the shift, a property of modality gaps specifically, or these particular datasets.

Two attempts to replicate outside dermatology failed for different reasons, and neither settles the question. On Camelyon17 the adversary head never trained: cross-entropy sat at $\ln 3$ from the first epoch at every λ_{adv} while a post-hoc probe on the same features read 0.94, despite a gradient graph verified correct (head cosine +1.000, encoder −1.000). On iWildCam the adversary did train — best-epoch cross-entropy 3.13 against $\ln 243 = 5.49$ — but the encoder never became invariant: the binary train-versus-OOD site probe moved only from 0.911 to 0.891 across $\lambda_{\text{adv}} \in [0, 10]$, against a floor of 0.5, and no detector collapsed. **We therefore report no result from either dataset, and make no claim that either resists invariance.** In both cases the manipulation did not take, so there was nothing to measure.

One observation survives these failures and may be useful to others. Invariance was obtained easily in the two-domain dermatology setting — leakage fell from 0.970 to 0.763 at $\lambda_{\text{adv}} = 0.25$ — and was not obtained at any weight in either multi-domain setting. Adversarial invariance may be substantially harder to achieve as the number of training domains grows, which would bound the settings in which the failure reported here can be observed at all. We offer this as an observation rather than a claim; it rests on two negative results.

Generality beyond dermatology remains untested.

Backbone coverage. Comparable invariance was not reached with a ResNet-50 backbone, so whether the inversion extends beyond EfficientNet-B3 is untested. Three of the nine runs failed their adversary gate outright and are reported as null experiments rather than negative results.

The objective comparison is bounded. MMD never reached the invariance level gradient reversal attains; it could only approach it by destroying the classifier. We therefore cannot say what moment-matching would do at matched invariance, only that at the leakage levels it reaches, gradient reversal has already inverted and MMD has not.

Detector non-independence. The five scoring rules are functions of the same 16-dimensional latent. The backbone analysis and the single-encoder replication provide the independent evidence.

Calibration arithmetic. Out-of-domain ECE is close to mean confidence minus accuracy in this setting; the confidence, accuracy and ECE results are one phenomenon viewed three ways, not three independent findings.

Seeds. Five seeds at the endpoints, three at interior points. Effect sizes are large relative to seed variance, but formal testing across seven points at $n = 3$ is not claimed.

Splits. ISIC splits are image-level, not patient-level; some patients may contribute images to both partitions. Because in-distribution accuracy is compared *across* λ_{adv} *under an identical split*, any inflation applies equally to every row and does not threaten the reported pattern, though absolute values may be optimistic. PAD splits are patient-level, disjoint by patient and by lesion.

Third-domain coverage. 3,887 of 16,577 Fitzpatrick17k images were obtained; the set is web-scraped under mixed copyright, used for evaluation only, and not redistributed.

7. Conclusion

Under domain-adversarial training, out-of-distribution detection does not merely weaken. It reverses: a model asked which inputs are unfamiliar answers with its own test data and ranks a new clinic’s images as the more routine of the two. The reversal appears at the smallest adversarial weight that achieves invariance, holds on a dataset the adversary never saw, reproduces without the factorized architecture, and does not appear under moment-matching objectives that reduce leakage along the same axis.

It is invisible to everything a developer monitors. In-distribution accuracy holds, in-distribution calibration is unchanged, and the domain leakage probe improves exactly as intended, while confidence on unseen domains overtakes confidence on familiar data and recall for a premalignant class falls to 0.016.

We could not explain the inversion after refuting six candidate accounts and a pre-registered synthetic reproduction. Explaining it is not required to act on it: reporting out-of-distribution detection, out-of-domain calibration and cross-domain accuracy as functions of achieved invariance costs almost nothing and makes the failure visible.

Data and code availability

<https://github.com/TranDucVinh-20225538/CSG-SKin>. ISIC 2019 CC BY-NC 4.0; PAD-UFES-20 CC BY 4.0; Fitzpatrick17k annotations CC BY-NC-SA 3.0 with images used for evaluation only and not redistributed.

Declaration of competing interest

The authors declare no competing interests.

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